Assignment - 4

Part-A: Feature Map Computation:

Input feature map size: 4×4×2

Convolution kernel size: 2×2×2

1. Regular Convolution (No Padding, Stride=2)

Output size = floor((N-F)/S)+1

Where, N = 4 (Input size)

F = 2 (Filter size)

S = 2 (stride)

Output Height = (4-2)/2+1 = 2

Output Width = (4-2)/2+1 = 2

Output depth = 1 (since one filter)

Final Output size = 2×2×1

2. Regular Convolution (With Zero Padding, Stride=2)

Assume padding P = 1

New input size = 4+2P = 4+2 = 6

Output size = (6-2)/2+1 = 3

Final output size: 3×3×1

3. Depthwise convolution - (No Padding, Stride=1)

Each input channel is convolved with its own filter.

No mixing between channels

So, Input channels = 2

Output channels = 2

Output size = (4-2)/1+1 = 3

Final output size = 3 × 3 × 2

Part - B:

```
import numpy as np
from scipy import signal
import matplotlib.pyplot as plt
from PIL import Image
img_path = 'home/saidul/Saidul_0.jpg'
img = Image.open(img_path).convert('L')
img_array = np.array(img, dtype=np.float32)/255.0
edge_kernel = np.array([
                [-1, 0, 1]
                [-2, 0, 2]
                [-1, 0, 1]
            ])
```

```
sharpen_kernel = np.array([[0,-1, 0],
                           [-1, 5,-1],
                           [0,-1, 0]])

blur_kernel = np.array([[1,2,1],
                        [2,4,2],
                        [1,2,1]], dtype=np.float32) / 16.0

kernels = { 'Edge Detection': edge_kernel,
            'Sharpening': sharpen_kernel,
            'Blurring': blur_kernel
          }

plt.figure(figsize=(12,10))

plt.subplot(2,2,1)
plt.imshow(img_array, cmap='gray')
plt.title('Original Face Image')
plt.axis('off')

for i,(name,kernel) in enumerate(kernels.items(),2):
    output = signal.convolve2d(img_array, kernel,
                               mode='same', boundary='symm')

    plt.subplot(2,2,i)
```

```
        plt.title(name)
        plt.axis('off')
    plt.tight_layout()
    plt.show()
```

Analysis: The three custom kernels demonstrate different effects on my face image. The edge detection kernel highlights boundaries and contours by emphasizing intensity changes, making facial features sharp and prominent. The sharpening kernel enhances fine details and edges, resulting in a crisper image with better definition around facial structures, though it can amplify noise. The blurring kernel smooths the image by averaging neighboring pixels, reducing noise and creating a softer appearance while losing fine details like skin texture. Overall, edge detection and sharpening are useful for feature extraction, while blurring is effective for noise reduction and preprocessing.

Part-c:

Spatially Separable kernels:

1. Horizontal Edge
2. Vertical Edge
3. Box Blur
4. Gaussian Blur
5. Prewitt Horizontal
6. Prewitt ~~Vertical~~
7. Simple Horizontal Line Detector
8. Simple Vertical Line Detector
9. Laplacian of Gaussian
10. Motion Blur

Spatially Non-Separable kernels:

1. Diagonal Edge Detector
2. Corner Detector
3. Circular Blob Detector
4. Cross Detector
5. Checkerboard Pattern
6. Arbitrary 3×3 random kernel
7. Sobel combined

8. Laplacian
9. Emboss Filter
10. Unsharp Mask

Part-D:

```
import tensorflow as tf

from tensorflow.keras.applications import MobileNetV2

import matplotlib.pyplot as plt

base_model = MobileNetV2(weights='imagenet', include_top=False)

layer = base_model.get_layer('Conv1')

filters = layer.get_weights()[0]

plt.figure(figsize=(10,10))

for i in range(16):
    plt.subplot(4,4,i+1)
    plt.imshow(filters[:,:,0,i], cmap='gray')
    plt.axis('off')
    plt.title(f'Filter {i+1}')
plt.suptitle('Learned kernels from First Conv Layer of
              MobileNetV2')
plt.show()
```

Discussion:

The first convolutional layer of MobileNetV2 learns a variety of basic level filters automatically during training on ImageNet. Visualization of the 3×3 filters shows several patterns resembling classical edge detectors, including vertical, horizontal, and diagonal gradients similar to Sobel operators. Some filters clearly act as edge detectors, while others appear to detect textures or blobs. This demonstrates that CNNs learn hierarchical features starting from basic edge detection in early layers which are then combined in deeper layers to form more complex representations such as eyes, faces or objects. The presence of Sobel-like patterns confirms that hand-crafted filters are naturally discovered by backpropagation, validating the biological plausibility of CNNs for visual feature extraction

Discussion:

The first convolutional layer of MobileNetV2 learns a variety of low-level filters automatically during training on ImageNet. Visualization of the 3×3 filters shows several patterns resembling classical edge detectors, including vertical, horizontal, and diagonal gradients similar to Sobel operators. Some filters clearly act as edge detectors, while others appear to detect textures or blobs. This demonstrates that CNNs learn hierarchical features starting from basic edge detection in early layers, which are then combined in deeper layers to form more complex representations such as eyes, faces or objects.

The presence of Sobel-like patterns confirms that hand-crafted filters are naturally discovered by backpropagation, validating the biological plausibility of CNNs for visual feature extraction

Part A: Feature Map Computation

- Input feature map size: 4 × 4 × 2
- Convolution kernel size: 2 × 2 × 2

1. Regular Convolution (No Padding, Stride = 2)

Formula: Output size = floor((N − F) / S) + 1

Where:

- N = 4 (input size)
- F = 2 (filter size)
- S = 2 (stride)

Output height = (4 − 2) / 2 + 1 = 2 Output width = (4 − 2) / 2 + 1 = 2

Since one filter is used, output depth = 1

Final output size: 2 × 2 × 1

Intermediate Computation

Stride = 2, so filter moves to positions:

- (0,0), (0,2)
- (2,0), (2,2)

Each output value is computed as:

O(i, j) = sum over channels and kernel elements = Σ (Input × Kernel)

Each involves element-wise multiplication of a 2×2×2 block with the kernel, followed by summation.

Final Feature Map Size: 2 × 2 × 1

2. Regular Convolution (With Zero Padding, Stride = 2)

Assume padding P = 1

New input size = 4 + 2P = 4 + 2 = 6

Output size = (6 − 2) / 2 + 1 = 3

Final output size: 3 × 3 × 1

Intermediate Computation

After padding, input becomes 6 × 6 × 2

Filter moves to positions:

- (0,0), (0,2), (0,4)
- (2,0), (2,2), (2,4)
- (4,0), (4,2), (4,4)

Each output value is computed as:

$O(i, j) = \Sigma (Input\_padded \times Kernel)$

At borders, some values in the input are zeros due to padding.

Final Feature Map Size: 3 × 3 × 1

3. Depthwise Convolution (No Padding, Stride = 1)

- Each input channel is convolved with its own filter
- No mixing between channels

So:

- Input channels = 2
- Output channels = 2

Output Dimension Calculation

Output size = (4 − 2) / 1 + 1 = 3

Final output size: 3 × 3 × 2

Intermediate Computation

For each channel separately:

Channel 1:

$O1(i, j) = \Sigma (Input\_channel1 \times Kernel\_channel1)$

Channel 2:

$O2(i, j) = \Sigma (Input\_channel2 \times Kernel\_channel2)$

Each uses a 2 × 2 filter on its respective channel.

Final Feature Map

Two separate feature maps of size 3 × 3:

Channel 1:

[O1(0,0) O1(0,1) O1(0,2)] [O1(1,0) O1(1,1) O1(1,2)] [O1(2,0) O1(2,1) O1(2,2)]

Channel 2:

[O2(0,0) O2(0,1) O2(0,2)] [O2(1,0) O2(1,1) O2(1,2)] [O2(2,0) O2(2,1) O2(2,2)]

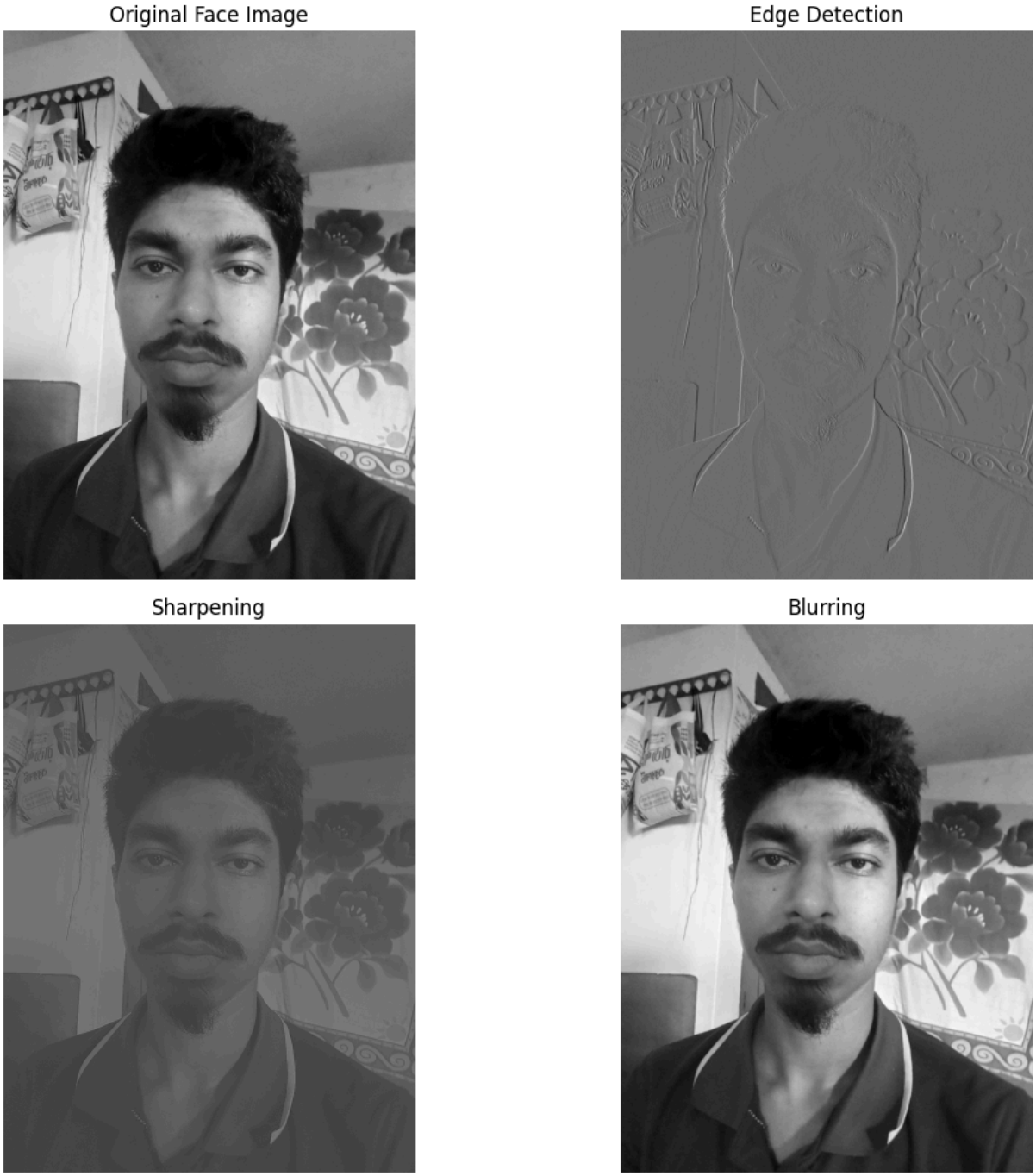
Summary

1. Regular convolution (no padding, stride 2): 2 × 2 × 1
2. Regular convolution (padding = 1, stride 2): 3 × 3 × 1
3. Depthwise convolution (no padding, stride 1): 3 × 3 × 2

In [3]:

```
import numpy as np
from scipy import signal
import matplotlib.pyplot as plt
from PIL import Image

# Load one of your face images (grayscale for simplicity)
img_path = '/home/saidul/Desktop/4-2/Deep Learning/Assignment1/data/faces/pe
img = Image.open(img_path).convert('L')
img_array = np.array(img, dtype=np.float32) / 255.0

# 1. Edge Detection Kernel (Sobel-like)
edge_kernel = np.array([
    [-1, 0, 1],
    [-2, 0, 2],
    [-1, 0, 1]
])

# 2. Sharpening Kernel
sharpen_kernel = np.array([[ 0, -1,  0],
                           [-1,  5, -1],
                           [ 0, -1,  0]])

# 3. Blurring Kernel (Gaussian approximation)
blur_kernel = np.array([[1, 2, 1],
                        [2, 4, 2],
                        [1, 2, 1]], dtype=np.float32) / 16.0

kernels = {
    'Edge Detection': edge_kernel,
    'Sharpening': sharpen_kernel,
    'Blurring': blur_kernel
}

# Apply and visualize
plt.figure(figsize=(12, 10))

plt.subplot(2, 2, 1)
plt.imshow(img_array, cmap='gray')
plt.title('Original Face Image')
plt.axis('off')
```

```
for i, (name, kernel) in enumerate(kernels.items(), 2):
    output = signal.convolve2d(img_array, kernel, mode='same', boundary='sym
    plt.subplot(2, 2, i)
    plt.imshow(output, cmap='gray')
    plt.title(name)
    plt.axis('off')

plt.tight_layout()
plt.show()
```



Part C: Spatial Separability Analysis

Spatially Separable Kernels (10 examples):

1. Horizontal Edge (Sobel X)
2. Vertical Edge (Sobel Y)
3. Box Blur / Average Filter
4. Gaussian Blur (any size)
5. Prewitt Horizontal
6. Prewitt Vertical
7. Simple Horizontal Line Detector
8. Simple Vertical Line Detector
9. Laplacian of Gaussian (approximated separable)
10. Motion Blur (horizontal or vertical)

Spatially Non-Separable Kernels (10 examples):

1. Diagonal Edge Detector
2. Corner Detector
3. Circular Blob Detector
4. Cross (plus sign) Detector
5. Checkerboard Pattern
6. Arbitrary 3×3 random kernel
7. Sobel combined (magnitude)
8. Laplacian (standard 3×3)
9. Emboss Filter
10. Unsharp Mask (complex)

Note: A kernel is spatially separable if it can be written as the outer product of a column vector and a row vector (e.g., Sobel vertical = [-1,0,1]^T × [1,2,1]).

In [4]:
```
import tensorflow as tf
from tensorflow.keras.applications import MobileNetV2
import matplotlib.pyplot as plt

# Load pre-trained model
base_model = MobileNetV2(weights='imagenet', include_top=False)

# Get first convolutional layer (Conv1)
layer = base_model.get_layer('Conv1')
filters = layer.get_weights()[0]  # Shape: (3, 3, 3, 32)

# Visualize first 16 filters (first input channel)
plt.figure(figsize=(10, 10))
for i in range(16):
    plt.subplot(4, 4, i+1)
    plt.imshow(filters[:, :, 0, i], cmap='gray')
    plt.axis('off')
    plt.title(f'Filter {i+1}')
plt.suptitle('Learned Kernels from First Conv Layer of MobileNetV2')
plt.show()
```

```
2026-04-26 22:52:32.226894: I external/local_tsl/tsl/cuda/cudart_stub.cc:32] Could not find cuda drivers on your machine, GPU will not be used.
2026-04-26 22:52:32.299710: I external/local_tsl/tsl/cuda/cudart_stub.cc:32] Could not find cuda drivers on your machine, GPU will not be used.
2026-04-26 22:52:32.396060: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:479] Unable to register cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been registered
2026-04-26 22:52:32.484174: E external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:10575] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been registered
2026-04-26 22:52:32.484645: E external/local_xla/xla/stream_executor/cuda/cuda_blas.cc:1442] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already been registered
2026-04-26 22:52:32.593705: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
2026-04-26 22:52:33.881613: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not find TensorRT
/tmp/ipykernel_73439/2045871189.py:6: UserWarning: `input_shape` is undefined or non-square, or `rows` is not in [96, 128, 160, 192, 224]. Weights for input shape (224, 224) will be loaded as the default.
  base_model = MobileNetV2(weights='imagenet', include_top=False)
```

Learned Kernels from First Conv Layer of MobileNetV2